

Math 465: Introduction to Combinatorics

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Quiz on Thursday, September 10 in class

Homework #1 due tonight! Homework #2 will be posted soon.

New office hours! Tue 1:15-2:15, Wed 4:30-5:30, Fri 1:30-2:30

These slides will be posted on Canvas.

Binomial identities (1)

There are many algebraic identities involving binomial coefficients. Behind each of them, there is a combinatorial correspondence.

To prove an identity, we either (a) “reverse engineer” an enumerative problem that can be solved in two different ways, or (b) establish a bijection between two sets, then enumerate each of them.

Proposition

$$\binom{n}{k} = \binom{n}{n-k}$$

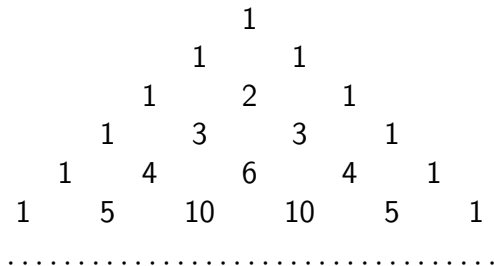
Proof

Set up a bijection between k -element and $(n-k)$ -element subsets.

Binomial identities (2)

Proposition

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$



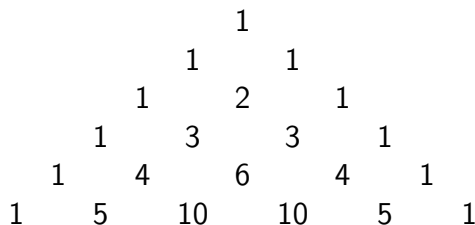
Proof

Count subsets of an n -element set, according to their size.

Binomial identities (3)

Proposition

$$\binom{n}{0} - \binom{n}{1} + \cdots + (-1)^n \binom{n}{n} = 0$$



Proof

Get a bijection between subsets of even and odd size, by matching each subset containing a fixed element s to a subset not containing s .

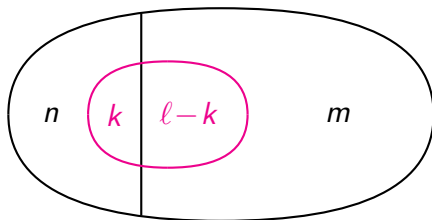
Binomial identities (4)

Theorem

$$\sum_k \binom{n}{k} \binom{m}{\ell - k} = \binom{n+m}{\ell}$$

Proof

Consider a disjoint union of two sets of sizes n and m , respectively. Then count the ℓ -element subsets of this union.



Binomial identities (5)

Corollary

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$$

Proof #1

Apply the identity

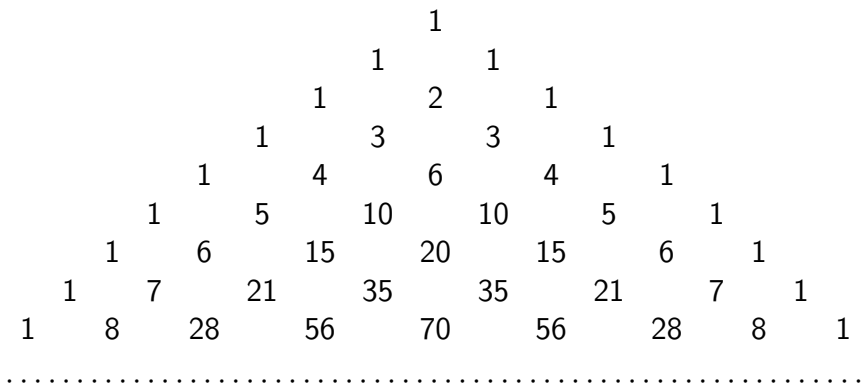
$$\sum_k \binom{n}{k} \binom{m}{\ell - k} = \binom{n+m}{\ell}$$

with $\ell = m = n$, and use that $\binom{n}{n-k} = \binom{n}{k}$.

Binomial identities (6)

Example

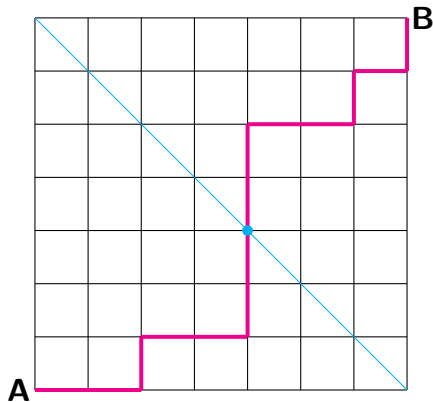
$$1^2 + 4^2 + 6^2 + 4^2 + 1^2 = 70 = \binom{8}{4}$$



Binomial identities (7)

Proof #2 of the identity $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$

The number of **lattice paths** from **A** = (0, 0) to **B** = (n, n) equals $\binom{2n}{n}$.
For each of these paths, look where it crosses **the diagonal**.



Another binomial identity

Theorem

$$\sum_k k \binom{n}{k} = n 2^{n-1}$$

Example: $n = 5$

$$0 \cdot 1 + 1 \cdot 5 + 2 \cdot 10 + 3 \cdot 10 + 4 \cdot 5 + 5 \cdot 1 = 80 = 5 \cdot 2^4.$$

Proof

Suppose we need to choose a **subset** of an n -element set, then pick an element of this subset (e.g., a **committee** and its chairperson). The number of such choices can be computed in two different ways:

- choose a k -element subset, then pick one its k elements;
- pick one of the n elements, then add a subset of the rest.

Binomial Theorem

Finally, an explanation of the term “binomial coefficients:”

$$(x + y)^0 = 1$$

$$(x + y)^1 = x + y$$

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

$$(x + y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$$

.....

Binomial Theorem

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

Proof of the Binomial Theorem

Binomial Theorem

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

Proof

Expand $(x + y)^n$ as the sum of n -letter words in the alphabet $\{x, y\}$:

$$\begin{aligned}(x + y)^3 &= (x + y)(x + y)(x + y) \\ &= xxx + xxy + xyx + xyy + yxx + yxy + yyx + yyy.\end{aligned}$$

(Here we view each word as a monomial in x and y .) Collect terms.
The coefficient of $x^k y^{n-k}$ is $\binom{n}{k}$ [why?].

This identity has an interpretation in probability theory (for $x+y=1$).

Binomial identities via the Binomial Theorem (1)

Many binomial identities can be proved using the Binomial Theorem

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Example

Set $x = y = 1$ to get

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

Example

Set $x = -1$ and $y = 1$ to get

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0.$$

Binomial identities via the Binomial Theorem (2)

Example

Start with

$$(x + 1)^n \cdot (x + 1)^m = (x + 1)^{n+m}.$$

Use the Binomial Theorem to rewrite it as

$$\sum_{k=0}^n \binom{n}{k} x^k \cdot \sum_{j=0}^m \binom{m}{j} x^j = \sum_{\ell=0}^{n+m} \binom{n+m}{\ell} x^{\ell}.$$

Take the coefficient of x^{ℓ} on both sides to get

$$\sum_k \binom{n}{k} \binom{m}{\ell - k} = \binom{n+m}{\ell}.$$

Binomial identities via the Binomial Theorem (3)

Example

Differentiate

$$(x + 1)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

to get

$$n(x + 1)^{n-1} = \sum_{k=1}^n \binom{n}{k} kx^{k-1}.$$

Now set $x = 1$ to obtain

$$n2^{n-1} = \sum_k k \binom{n}{k}.$$

Multinomial theorem

Recall that the multinomial coefficient

$$\binom{n}{n_1 \dots n_j} \stackrel{\text{def}}{=} \frac{n!}{n_1! \dots n_j!}$$

is the number of words in the alphabet $\{x_1, \dots, x_j\}$ that consist of n_1 copies of letter x_1 , $\dots$, n_j copies of letter x_j . Here $n = n_1 + \dots + n_j$.

Multinomial theorem

$$(x_1 + \dots + x_j)^n = \sum_{n_1 + \dots + n_j = n} \binom{n}{n_1 \dots n_j} x_1^{n_1} \dots x_j^{n_j}.$$

Proof

Expand $(x_1 + \dots + x_j)^n$ as the sum of all words of length n in the alphabet $\{x_1, \dots, x_j\}$, viewing words as monomials. Collect terms.

Multinomial theorem: Example

$$\begin{aligned}(a + b + c + d)^3 &= (a + b + c + d)(a + b + c + d)(a + b + c + d) \\&= aaa + aab + aac + aad + aba + abb + abc + abd \\&\quad +aca + acb + acc + acd + ada + adb + adc + add \\&\quad +baa + bab + bac + bad + bba + bbb + bbc + bbd \\&\quad +bca + bcb + bcc + bcd + bda + bdb + bdc + bdd \\&\quad +caa + cab + cac + cad + cba + cbb + cbc + cbd \\&\quad +cca + ccb + ccc + ccd + cda + cdb + cdc + cdd \\&\quad +daa + dab + dac + dad + dba + dbb + dbc + dbd \\&\quad +dca + dcb + dcc + dcd + dda + ddb + ddc + ddd \\&= a^3 + b^3 + c^3 + d^3 + 6abc + 6abd + 6acd + 6bcd \\&\quad + 3a^2b + 3ab^2 + 3a^2c + 3ac^2 + 3a^2d + 3ad^2 \\&\quad + 3b^2c + 3bc^2 + 3b^2d + 3bd^2 + 3c^2d + 3cd^2\end{aligned}$$

Multinomial theorem, continued

Problem

What is the largest coefficient of a monomial in the expansion of the polynomial $(x + y + z)^6$?

Solution

For $a + b + c = 6$, the largest value of

$$\binom{6}{a \ b \ c} = \frac{6!}{a! \ b! \ c!}$$

is attained at $a = b = c = 2$ [why?], so the answer is $\frac{6!}{2^3} = 90$.

Generalized binomial coefficients

Definition

For α a **complex number** and k a nonnegative integer, the **generalized binomial coefficient** $\binom{\alpha}{k}$ is defined by

$$\binom{\alpha}{k} \stackrel{\text{def}}{=} \frac{\alpha(\alpha - 1) \cdots (\alpha - k + 1)}{k!}.$$

When $\alpha = n$ is a nonnegative integer, we recover the usual definition:

$$\binom{n}{k} = \frac{n(n - 1) \cdots (n - k + 1)}{k!} = \frac{n!}{(n - k)!k!}.$$

Note that for $k > n$, we get $\binom{n}{k} = 0$.

Generalized binomial coefficients: the case $\alpha = -n$

The generalized binomial coefficients are defined by

$$\binom{\alpha}{k} \stackrel{\text{def}}{=} \frac{\alpha(\alpha - 1) \cdots (\alpha - k + 1)}{k!}.$$

Example: $\alpha = -n$

$$\begin{aligned}\binom{-n}{k} &= \frac{(-n)(-n-1)\cdots(-n-k+1)}{k!} \\ &= (-1)^k \frac{(n+k-1)\cdots(n+1)n}{k!} \\ &= (-1)^k \binom{n+k-1}{k}.\end{aligned}$$

Generalized Binomial Theorem

Meta-Theorem [I. Newton]

The Binomial Theorem identity

$$(x + y)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k y^{\alpha-k}$$

holds whenever it makes sense.

In particular, it holds

- when α is an integer (possibly negative), or
- when $x + y$ and y are positive real numbers

(assuming that the infinite sum on the right-hand side converges, or we work in the realm of *formal power series*).

Proof of Newton's generalized Binomial Theorem

Viewing $(x + y)^\alpha$ as a function of x , apply Taylor's formula:

$$f(x) = (x + y)^\alpha,$$

$$f'(x) = \alpha(x + y)^{\alpha-1},$$

$$f''(x) = \alpha(\alpha - 1)(x + y)^{\alpha-2},$$

.....

$$f^{(k)}(x) = \alpha(\alpha - 1) \cdots (\alpha - k + 1)(x + y)^{\alpha-k},$$

.....

$$\begin{aligned} f(x) = (x + y)^\alpha &= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k \\ &= \sum_k \frac{\alpha(\alpha - 1) \cdots (\alpha - k + 1) y^{\alpha-k}}{k!} x^k \\ &= \sum_k \binom{\alpha}{k} x^k y^{\alpha-k}. \end{aligned}$$

Binomial Theorem for negative powers

Setting $\alpha = -n$ in Newton's theorem, we get:

$$(1+x)^{-n} = \sum_{k=0}^{\infty} \binom{-n}{k} x^k = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{k} x^k = \sum_{k=0}^{\infty} (-1)^k \binom{n+k-1}{n-1} x^k.$$

Corollary

$$(1-x)^{-n} = \sum_{k=0}^{\infty} \binom{n+k-1}{n-1} x^k.$$

Examples

$$(1-x)^{-1} = \sum_k \binom{k}{0} x^k = \sum_k x^k,$$

$$(1-x)^{-2} = \sum_k \binom{k+1}{1} x^k = \sum_k (k+1) x^k.$$